Koopman-von Neumann Mechanics on the Baroclinic Annulus

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Table of Contents

1. Background
   - Koopman-von Neumann mechanics
   - Modeling the baroclinic annulus experiment
2. Framework
   - Prescribing a Velocity Field
   - Derivation of KvN operator
3. Numerical Simulation
   - Galerkin projection and matrix representation
   - Choosing Basis Functions
   - Wavefunction approximation and evolution

KvN and Continuity

Given a dynamical system of the form,

$$\dot{\mathbf{x}}(t) = F(\mathbf{x})$$
$$x_0 = \mathbf{x}(0)$$

where F is a vector field, the **Koopman-von Neumann equation** is

$$\partial_t \psi = \mathcal{L}^*_{KvN}\psi := -F \cdot \nabla \psi - \frac{1}{2}\mathrm{div}(F)\psi$$

where ψ is a wavefunction solving this equation. Then $\rho := |\psi|^2$ is a density which *automatically satisfies* the **continuity equation**:

$$\partial_t \rho + \nabla \cdot (\rho v) = 0$$

where (in the context of fluid dynamics $\rho(x, t)$ is the mass density of the fluid and $v(x, t)$ is the velocity vector field.
In [7], it was shown that we can apply this KvN framework, on bounded domains, which motivates the following example.

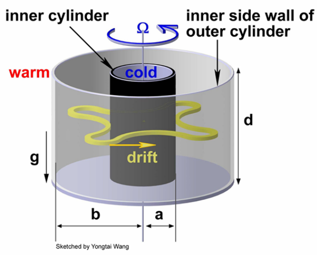
The Baroclinic Annulus

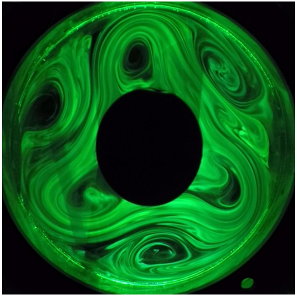
The **baroclinic annulus** is a classical experiment in fluid and thermodynamics. In this project we consider a 2D model described on the domain;

Definition

We consider the 2D model of the baroclinic annulus with inner radius r_0 and outer radius r_1, rotating with an angular speed ω_0, defining our domain:

$$\Omega := \{(r, \theta) \in \mathbb{R}^2 : r_0 < r < r_1,\ 0 \leq \theta < 2\pi\}$$





Images from [1].

Kinematic model

On this domain, we model the dynamics of the experiment by prescribing a continuous velocity vector field $V : \bar{\Omega} \times \mathbb{R} \to \mathbb{R}^2$ of the form

$$V(r, \theta, t) := \begin{bmatrix} V_r(r, \theta, t) \\ V_\theta(r, \theta, t) \end{bmatrix}$$

satisfying homogeneous boundary conditions, $V_r(r_0, \theta, t) = 0 = V_r(r_1, \theta, t)$.
In order to recreate the dynamics observed experimentally, we work with a toy model using the velocity vector field;

$$V_r(r, \theta, t) := \sum_{m=1}^{N} A_m(r) \sin(m\theta - \omega_m t + \phi_m^{(r)})$$

$$V_\theta(r, \theta, t) := \omega_0 r + \sum_{m=1}^{N} B_m(r) \cos(m\theta - \omega_m t + \phi_m^{(\theta)})$$

where $A_m(r)$ and $B_m(r)$ are smooth scalar-valued functions, $m \in \mathbb{N}$ is the wavenumber, and $\phi_m^{(r)}$ and $\phi_m^{(\theta)}$ are just phase shift constants. Note that importantly, this vector field is not necessarily divergence free, and so this models a *compressible fluid*.

The KvN Operator

Rewriting the continuity equation in terms of our chosen velocity vector field inpolar coordinates yields:

$$\partial_t \rho + \frac{1}{r}\partial_r(r\rho V_r) + \frac{1}{r}\partial_\theta(\rho V_\theta) = 0$$

To derive the KvN operator for this equation, we consider a $\psi(t,r,\theta) \in L^2(\Omega,\mathbb{C})$ satisfying $\rho = |\psi|^2 = \psi\psi^*$. Substituting this into the continuity equation and performing some derivations, we get:

$$0 = \partial_t\psi + V_r\partial_r\psi + \frac{1}{r}V_\theta\partial_\theta + \frac{1}{2}\psi(\nabla \cdot V)$$

Then we can define the KvN operator in polar coordinates as

$$\mathcal{Q} := -V_r\partial_r - \frac{1}{r}V_\theta\partial_\theta - \frac{1}{2}(\nabla \cdot V)$$

which yields the KvN equation, $\partial_t\psi = \mathcal{Q}\psi$. Furthermore, we also have that $\mathcal{Q}$ is *skew-symmetric*.

Approximating the KvN Operator

To approximate the Koopman–von Neumann operator numerically, we use the **Galerkin projection** as outlined in [4] and [5].

1. Choose a suitable set of basis functions $\{\phi_k\}_{k=1}^{N}$ on the domain Ω.
2. Compute the Gram matrix $G_{ij} = \langle \phi_i, \phi_j \rangle$ and the action matrix $A_{ij} = \langle \phi_i, \mathcal{Q}\phi_j \rangle$.
3. Perform a basis whitening transformation via $G^{-1/2}$ to obtain the skew-Hermitian matrix representation:

$$Q = \frac{1}{2} G^{-1/2}(A - A^{\dagger})G^{-1/2}$$

Under this transformation, the discrete generator Q is strictly skew-Hermitian ($Q^{\dagger} = -Q$), ensuring that its propagator $U(t) = \exp(tQ)$ is unitary. The eigenvalues and eigenvectors of Q yield the spectral properties of the discretized system.

Basis functions

In our case, we need basis functions $\phi_k(r, \theta)$ on Ω satisfying

1. **Homogeneous Dirichlet boundary conditions**:
   $\phi_k(r_0, \theta) = 0 = \phi_k(r_1, \theta), \quad \forall \theta \in [0, 2\pi)$, and
2. **Periodic boundary condition**: $\phi_k(r, \theta) = \phi_k(r, \theta + 2\pi), \quad \forall r \in (r_0, r_1)$.

Thus we consider basis functions of the form

$$\phi_{n,m}(r, \theta) = R_n(r) \cdot \Theta_m(\theta)$$

where $R(r)$ enforces the Dirichlet BCs and $\Theta(\theta)$ enforces the periodic BC. For this project, we experimented with using *Chebyshev polynomials* for the radial component and a *Fourier basis* for the angular component, i.e.

$$R_n(r) = (r - r_0)(r_1 - r)T_n(x(r)), \quad \Theta_m(\theta) = e^{i\theta m}$$

where $n = 0, 1, 2, \ldots$, and

$$x(r) = 2\left(\frac{r - r_0}{r_1 - r_0}\right) - 1$$

Resulting Wavefunctions

From the projected KvN operator, we can compute the wavefunction $\psi(r, \theta, t)$ associated with fluid density $\rho = |\psi|^2$, and evolve it in time.

1. Fix an initial distribution (we choose a radial Gaussian distribution on the annulus);

$$\psi_0(r, \theta) = \exp\left\{ \frac{-[r - \frac{1}{2}(r_0 + r_1)]^2}{\frac{1}{8}(r_1 - r_0)^2} \right\}$$

2. Project this initial wavefunction onto our chosen basis and compute the associated coefficients by solving the linear system,

$$GC_0 = \langle \phi_{n,m}, \psi_0 \rangle_{L^2}$$

for the initial coefficients C_0.

3. Evolve these coefficients in time using the propagator $C(t) = e^{tQ} C_0$ to define our wavefunction,

$$\psi(r, \theta, t) := \sum_{n,m} c_{n,m}(t) \phi_{n,m}(r, \theta)$$

where $c_{n,m}(t)$ are the entries in the coefficient matrix $C(t)$.

Quantum Algorithm Implementation

To test the feasibility of implementing this on a quantum computer, we perform the following quantum algorithm:

1. **Project** the matrix Q onto a quantum register (such that the Hilbert space dimension $2^{n_q} \geq N(2M+1)$, the dimension of the Galerkin basis). We then pad Q with zeros to match the dimensions, and reinforcee anti-Hermiticity by taking $Q = \frac{1}{2}(Q_{\text{pad}} - Q_{\text{pad}}^{\dagger})$
2. **Decompose** Q into the *Pauli operator basis* $\mathcal{P}_{n_q} = \{I, X, Y, Z\}^{\otimes n_q}$ to get
$$Q = \sum_k c_k P_k, \quad P_k \in \mathcal{P}_{n_q}$$
where $c_k \in i\mathbb{R}$ are imaginary coefficients.
3. **Evolve** the quantum state vector using the *2nd order symmetric Trotter Suzuki product expansion*:
$$U(\Delta t) = \left(\prod_{k=1}^{S} \exp\left(c_k P_k \frac{\Delta t}{2}\right)\right)\left(\prod_{k=S}^{1} \exp\left(c_k P_k \frac{\Delta t}{2}\right)\right)$$

Procedure based on [2], [3], and [6].

Test Case

Example: For $N = 10$, $M = 5$, we have a 110 dimensional basis, and thus we work with $n_q = 7$ qubits, requiring a $2^7 = 128$ quantum register.
With 100 time steps where $\Delta t = 0.01$ and 6264 Pauli strings we observed:

- The wavefunction norm $\|\psi\|_2 = 1$ is conserved, confirming exact unitarity.
- The quantum state fidelity $F = |\langle\psi_{\text{exact}}(1.0)|\psi_{\text{Trotter}}(1.0)\rangle|^2 \approx 0.9983$.

This confirms that storing $N(2M + 1)$ dimensional Galerkin system requires only $n_q = \lceil \log_2 N(2M + 1) \rceil$ qubits, providing an exponential compression in memory state space.

This project acts as a *proof of concept* as an application of the **KvN framework** on bounded domains, and in particular to classical fluid dynamics problems such as the baroclinic annulus experiment.

References

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